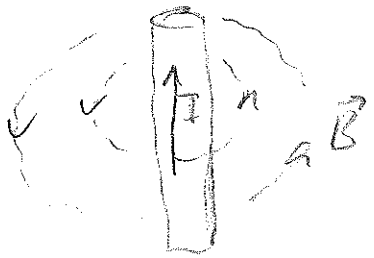
Practice Problems
Calculus Physics II

Name: Key **Date:** _____

Chapter 47 Current and the Magnetic Field

1) Magnetic field lines about a current-carrying wire



A) circle the wire in closed loops.
B) extend radially from the wire.
C) Choices A and B are both correct.
D) None of the above choices are correct.

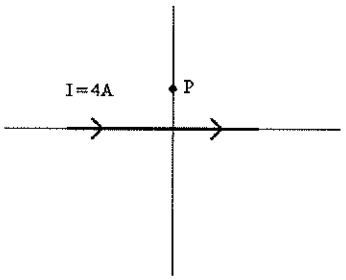
2) Two long parallel wires are placed side-by-side on a horizontal table. If the wires carry current in the same direction,

A) one wire is lifted slightly while the other wire is forced downward against the table's surface.
B) both wires are lifted slightly.
C) the wires pull toward each other.
D) the wires push away from each other.

3) A long straight wire carrying a 4-A current is placed along the x-axis as shown in the figure. What is the direction of the magnetic field at a point P, located at $y = 0.02$ m, due to this wire?

RHR #2

A) into the plane of the page
B) along the $+x$-axis
C) along the $-x$-axis
D) along the $+y$-axis
E) out of the plane of the page



4) A long straight wire carrying a 4-A current is placed along the x-axis as shown in the figure above. What is the magnitude and direction (into or out of the page) of the magnetic field at a point P, located at $y = 0.02$ m, due to the current in this wire? ($\mu_o = 1.257 \times 10^{-6}$ Tm/A)

$I = 4A$

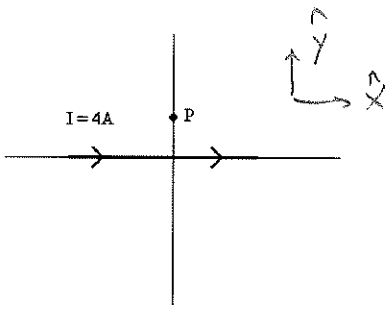
$r = 0.02\,m$

μ_0

$|\vec{B}| = \frac{\mu_0 I}{2\pi r} = 4 \times 10^{-5}\,T$

out of page

$\vec{B} = (4 \times 10^{-5}\,T)\,\hat{z}$



5) A long wire carrying a 2.0-A current is placed along the *y*-axis. What is the magnitude of the magnetic field at a point 0.60 m along the *x*-axis? ($\mu_o = 1.257 \times 10^{-6}$ Tm/A)

$|\vec{B}| = \frac{\mu_0 I}{2\pi r} = 6.67 \times 10^{-7}$ T into page

6) Two long parallel wires carry currents of 20 A and 5.0 A in the same directions. The wires are separated by 0.20 m. What is the magnetic field midway between the two wires? ($\mu_o = 1.257 \times 10^{-6}$ Tm/A) Assume out of the page is the positive direction.

See attached

$\Sigma B_z = -3 \times 10^{-5}$ T into page

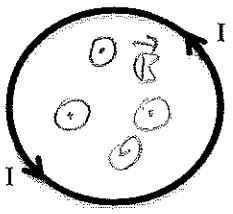
7) Two long parallel wires that are 0.40 m apart carry currents of 10 A in opposite directions. What is the magnetic field strength in the plane of the wires at a point that is 20 cm from one wire and 60 cm from the other? ($\mu_o = 1.257 \times 10^{-6}$ Tm/A) Assume out of the page is the positive direction.

See attached

$\Sigma B_z = -6.67 \times 10^{-6}$ T

8) A current-carrying loop of wire lies flat on a horizontal tabletop. When viewed from above, the current moves around the loop in a counterclockwise sense. For points on the tabletop *inside* the loop, the magnetic field lines caused by this current

A) circle the loop in a clockwise direction.
B) circle the loop in a counterclockwise direction.
C) points out of page.
D) point into page.



9) What is the magnetic field at the center of a circular loop of wire of radius 4.0 cm when a current of 2.0 A flows in the wire? ($\mu_o = 1.257 \times 10^{-6}$ Tm/A)

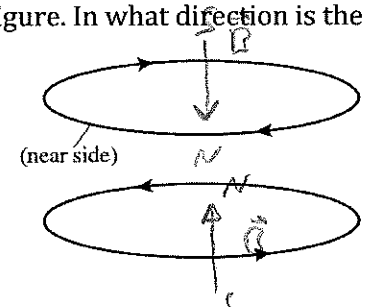
$r = 0.04$m
$I = 2$A
$N = 1$ loop

$|\vec{B}| = N\left(\frac{\mu_0 I}{2r}\right) = 3.14 \times 10^{-5}$ T out of page

10) A ring with a clockwise current (as viewed from above the ring) is situated with its center directly above another ring, which has a counterclockwise current, as shown in the figure. In what direction is the net magnetic force exerted on the top ring due to the bottom ring?

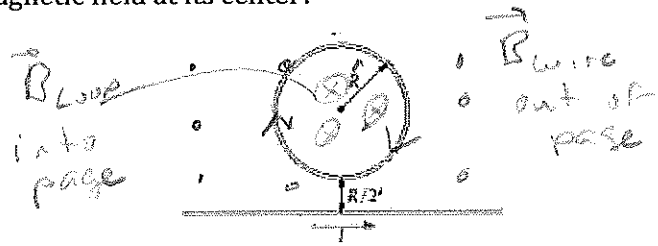
repel

A) upward
B) downward
C) to the left
D) to the right
E) The net force is zero.

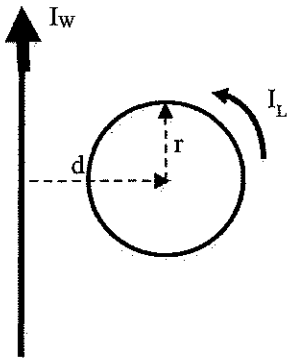


11) A single current-carrying circular loop of radius *R* is placed next to a long, straight wire, as shown in the sketch in the figure below. The current in the wire points to the right and is of magnitude *I*. In which direction must current flow in the loop to produce zero magnetic field at its center?

A) counterclockwise
B) clockwise
C) It is indifferent if clockwise or counterclockwise.
D) The current in the loop should be zero A.



12) A long current carrying wire is positioned d = 0.5 m from a current carrying loop of wire with a radius of r = 0.1 m. If the current in the wire is I_w = 5 A upwards and the current in the loop of wire is I_L = 0.8 A in the counterclockwise direction, determine the magnitude and direction of the net magnetic field strength at the center of the loop of wire.



$\Sigma \vec{B} = \vec{B}_w + \vec{B}_L$

$|\vec{B}_w| = \frac{\mu_0 I_w}{2\pi d} = 2\times10^{-6}\ T$ into page

$|\vec{B}_L| = \frac{\mu_0 I_L}{2r} = 5\times10^{-6}\ T$ out of page

$\Sigma B_z = (5\times10^{-6}\ T) + (-2\times10^{-6}\ T) = +3\times10^{-6}\ T$

13) Two long parallel wires carry currents of 5.0 A and 8.0 A in the opposite direction. The wires are separated by 0.30 m and have a length of L = 0.5 m. Find the magnitude of the magnetic force and direction (attract or repel) between the two wires. (μ_o = 1.257 x 10^{-6} Tm/A)

See attached

$|\vec{F}_m| = 1.33\times10^{-5}\ N$

repels

14) Two long parallel wires (L = 1.2 m) are 2 mm apart and carry currents of 40 A and 60 A in the same direction. What is the magnitude of the force that each wire exerts on the other wire? (μ_o = 1.257 x 10^{-6} Tm/A)

See attached

$|\vec{F}_m| = 0.29\,N$

attracts

15) How much current must pass through a 400-turn solenoid 4.0 cm long to generate a 1.0-T magnetic field at the center?

$|\vec{B}| = 1\,T$

$N = 400$ Loops

$L = 0.04\,m$

$|\vec{B}| = \frac{\mu_0 I N}{L}$

$I = \frac{BL}{\mu_0 N} = 79.6\,A$

16) A solenoid with 400 turns has a radius of 0.040 m and is 40 cm long. If this solenoid carries a current of 12 A, what is the magnitude of the magnetic field at the center of the solenoid if there is air inside the solenoid? $\mu_0 = 1.257 \times 10^{-6}$ Tm/A

$N = 400$ Loops

$L = 0.4\,m$

$I = 12A$

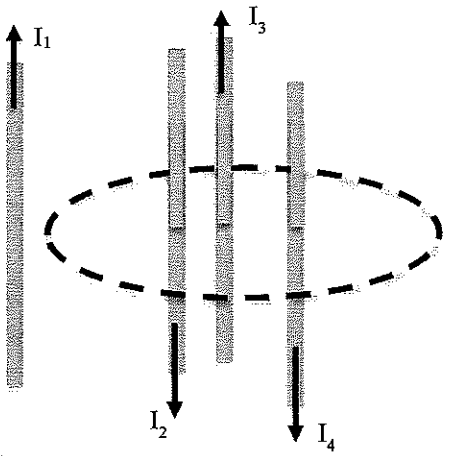
$|\vec{B}| = \frac{\mu_0 I N}{L} = 0.015\,T$

17) A solenoid with 400 turns has a radius of 0.040 m and is 40 cm long. If this solenoid carries a current of 12 A, what is the magnitude of the magnetic field at the center of the solenoid if the solenoid is filled with iron? (μ_{iron} = 6.5 x 10^{-3} Tm/A)

$|\vec{B}| = \frac{\mu_{Iron} I N}{L} = 78\,T$

18) Using Ampere's Law determine the Ampere's integral for the Amperian surface below.

$I_1 = 3$ A
$I_2 = 3$ A
$I_3 = 2$ A
$I_4 = 2$ A



$$\oint \vec{B} \cdot d\vec{L} = \mu_0 I_{in}$$

$$I_{in} = I_3 + (-I_2) + (-I_4)$$

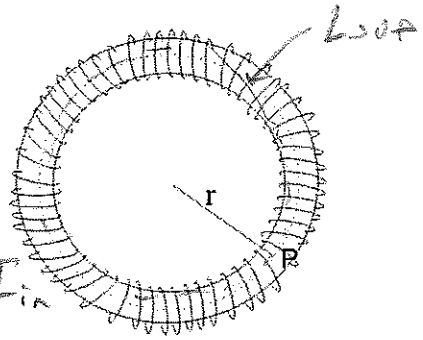
$$I_{in} = -3A$$

$$\oint \vec{B} \cdot d\vec{L} = \mu_0 (-3A)$$

$$= -3.77 \times 10^{-6} \, HA/m$$

19) Using Ampere's Law determine the magnitude of the magnetic field inside of a Toroid of N number of turns and a radius of R and current I.

A) Draw the Amperian loop and label the dL with direction.



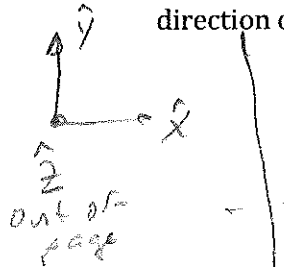
$$L = C = 2\pi r$$

B) Derive the equation for the magnetic field inside the toroid.

$$\oint \vec{B} \cdot dL = |\vec{B}| L = |\vec{B}| 2\pi r = \mu_0 I_{in}$$

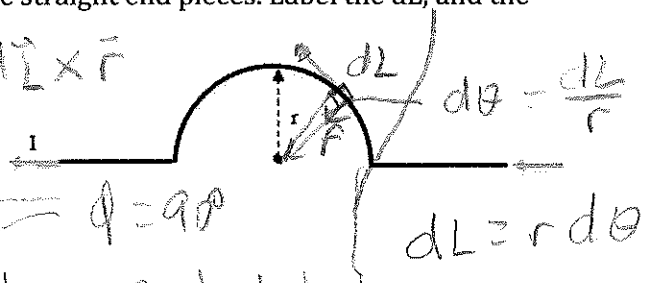
$$|\vec{B}| = \frac{\mu_0 I N}{2\pi r}$$

20) Using Biot-Savart Law, derive the equation for the magnetic field at point P, at the center of a half loop of current carrying wire. Consider only the loop of wire not the straight end pieces. Label the dL, and the direction of the magnetic field.



Direction of $\vec{B}$ given by $d\vec{L} \times \vec{r}$

out of page



$\phi = 90°$ angle between $d\vec{L}$ and $\vec{r}$

$$B_z = \frac{\mu_0 I}{4\pi} \int \frac{\sin\phi \, dL}{r^2}$$

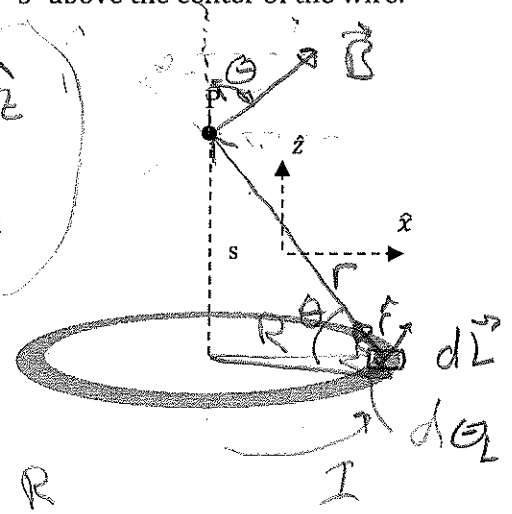
$$B_z = \frac{\mu_0 I}{4\pi r^2} \int_0^{\pi} r \, d\theta = \frac{\pi \mu_0 I}{4\pi r}$$

$$B_z = \frac{\mu_0 I}{4 r}$$

21) of radius R

For the circular loop of current carrying wire below, use Biot-Savart Law to derive the equation for the magnetic field along the z direction, direction at point P at a distance of "s" above the center of the wire. Label the dL, r and the magnetic field direction.

$\vec{B}$ - field perpendicular to both $\hat{r}$ and $d\vec{L}$



$$B_z = \vec{B}\cdot\hat{z} = |\vec{B}|\cos\theta$$

$$r = (s^2 + R^2)^{1/2} \qquad \cos\theta = \frac{R}{r}$$

$$\cos\theta = \frac{R}{(s^2+R^2)^{1/2}}$$

$$B_z = \frac{\mu_0 I}{4\pi}\int\left(\frac{\cos\theta}{r^2}\right) d\vec{L}\times\hat{r}$$

$$d\vec{L}\times\hat{r} = |d\vec{L}||\hat{r}|\sin\phi$$

$\hat{r}$ and $d\vec{L}$ are perpendicular $\phi = 90°$

$$B_z = \frac{\mu_0 I}{4\pi}\left[\left(\frac{R}{(s^2+R^2)^{1/2}}\right)\left(\frac{1}{(s^2+R^2)}\right)\right] d\vec{L}\times\hat{r}$$

$$B_z = \frac{\mu_0 I R}{4\pi(s^2+R^2)^{3/2}}\int R\, d\theta_L$$

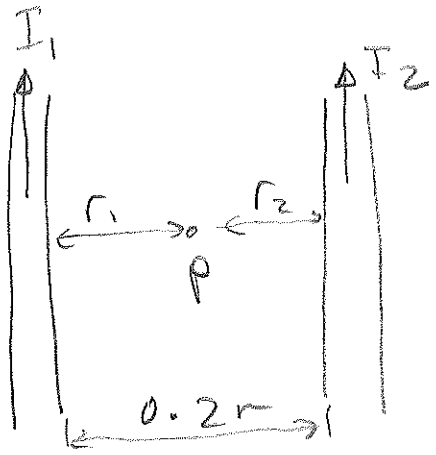
$$dL = ds = R\, d\theta_L$$

$$B_z = \frac{\mu_0 I R^2}{4\pi(s^2+R^2)^{3/2}}\int_0^{2\pi} d\theta_L = \frac{\mu_0 I R^2}{2(s^2+R^2)^{3/2}}$$

KEY

1) A
2) C
3) E
4) 4×10^{-5} T and out of page
5) 6.67×10^{-7} T and into the page
6) 3×10^{-5} T and into the page
7) 6.67×10^{-6} T and into the page
8) C
9) 3.14×10^{-5} T and out of page
10) A
11) B
12) 3.0×10^{-6} T and out of page
13) 1.33×10^{-5} N and repulsive
14) 0.29 N and attractive
15) 79.6 A
16) 0.015 T
17) 78 T
18) -3.8×10^{-6} HA/m
19) See solutions
20) See solutions
21) See solutions

6)



$I_1 = 20\,A$ $\quad I_2 = 5\,A$

$r_1 = 0.1m$ $\quad r_2 = 0.1m$

$\mu_0 = 1.257 \times 10^{-6}\,Tm/A$

$$|\vec{B}_1| = \frac{\mu_0 I_1}{2\pi r_1} = 4 \times 10^{-5}\,T \otimes$$

RHR #2 → into the page

$$|\vec{B}_2| = \frac{\mu_0 I_2}{2\pi r_2} = 1.0 \times 10^{-5}\,T$$

RHR #2 → out of page

$$\Sigma \vec{B} = \vec{B}_1 + \vec{B}_2$$

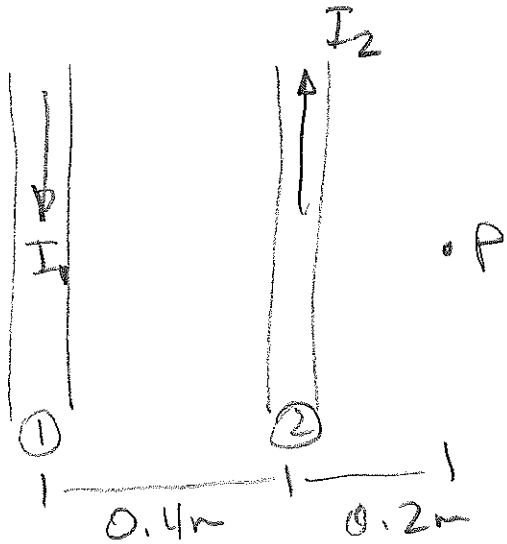
$$\Sigma B_z = (+1.0 \times 10^{-5}\,T) + (-4.0 \times 10^{-5}\,T)$$

(+) out of page; (−) ↑ into page

$$\Sigma B_z = -3 \times 10^{-5}\,T \text{ into page}$$

↑ into page

7)



$I_1 = 10A$

$r_1 = 0.6m$

$I_2 = 10A$

$r_2 = 0.2m$

$$\sum \vec{B} = \vec{B}_1 + \vec{B}_2$$

$$|\vec{B}_1| = \frac{\mu_0 I_1}{2\pi r_1} = 3.33 \times 10^{-6} T$$

out of page

$$\sum B_z = (+3.33 \times 10^{-6} T) + (-1.0 \times 10^{-5} T)$$

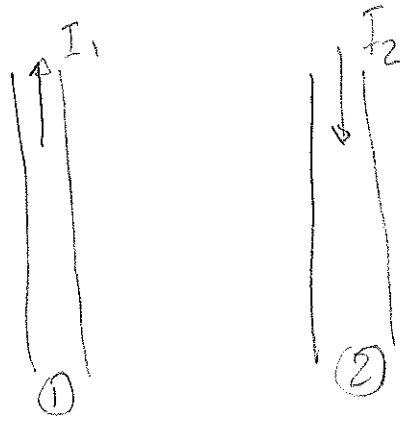
$$|\vec{B}_2| = \frac{\mu_0 I_2}{2\pi r_2} = 1.0 \times 10^{-5} T$$

into page

$$\sum B_z = -6.67 \times 10^{-6} T$$

↑ into page

13)



① need magnetic field from wire 1 at wire 2

$$|\vec{B}_1| = \frac{\mu_0 I_1}{2\pi r} = 3.3\times10^{-6}\ T$$

into page

$I_1 = 5A$

$r_1 = 0.3m$

② Force on wire 2 due to $|\vec{B}_1|$

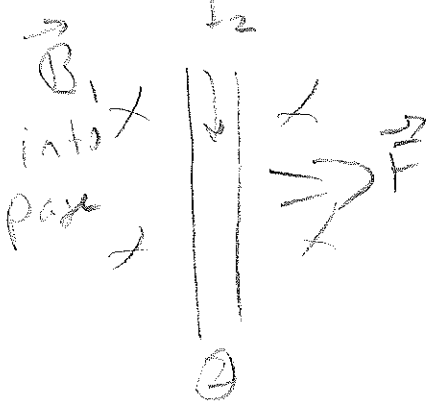
$$|\vec{F}_B| = I_2 L B_1$$

$I_2 = 8A$

$r = 0.5m$

$$|\vec{F}_B| = (8A)(0.5m)(3.3\times10^{-6}T)$$

$$|\vec{F}_B| = 1.33\times10^{-5}\ N$$

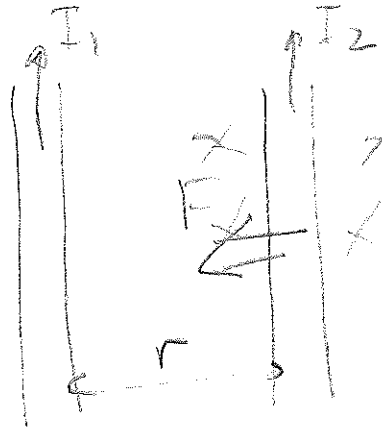


Right hand rule #1

Force directed to the right

Replusive

14)



① need $\vec{B}$ Field at wire 2 From wire 1

$$|\vec{B}_1| = \frac{\mu_0 I_1}{2\pi r} = 4 \times 10^{-3}\ T$$

into page

$I_1 = 40A$

$r_1 = 2 \times 10^{-3} m$

$L = 1.2 m$

$I_2 = 60A$

② need Force on wire 2 due to $|\vec{B}_1|$

$$|\vec{F}_m| = I_2 L B_1$$

$$|\vec{F}_m| = 0.29 N$$

→ attractive

Right hand rule # 1

Force to left

